

Question 1

Quadrilateral $BMKR$ has vertices at $B(6, 7)$, $M(9, 4)$, $K(6, 1)$, and $R(3, 4)$.

- Find the slopes, side lengths, and diagonal lengths of the quadrilateral.
- Name the properties used to classify the quadrilateral.
- Classify the quadrilateral.

Question 2

Quadrilateral $PHND$ has vertices at $P(7, -7)$, $H(5, -4)$, $N(5, -1)$, and $D(7, -4)$.

- Find the slopes, side lengths, and diagonal lengths of the quadrilateral.
- Name the properties used to classify the quadrilateral.
- Classify the quadrilateral.

Question 3

Quadrilateral $WHCS$ has vertices at $W(-9, 4)$, $H(-7, 5)$, and $C(-4, 4)$.

Point S		
$(-6, 2)$	$(-10, 1)$	$(-12, 3)$

- Determine the length of $\overline{WH}$ and $\overline{HC}$.
- Determine the slopes of $\overline{WH}$ and $\overline{HC}$.
- Find the slope of $\overline{CS}$ using each point S .
- At which coordinates is point S if quadrilateral $WHCS$ is an isosceles trapezoid?
- Verify that two sides of the trapezoid are congruent.

Question 4

Marissa and Carlos are classifying quadrilateral $MPFT$ with vertices at $M(-1, 9)$, $P(3, 6)$, $F(0, 3)$, and $T(-4, 6)$. Their classifications are shown.

Marissa's Classification	Carlos's Classification
Quadrilateral $MPFT$ is a parallelogram.	Quadrilateral $MPFT$ is a rectangle.

- Find the slopes, side lengths, and diagonal lengths of the quadrilateral.
- Name the properties used to classify the quadrilateral.
- Determine whose classification is correct.

Question 5

Quadrilateral $KCNR$ has vertices at $K(-8, -3)$, $C(-3, -1)$, $N(-5, -6)$, and $R(-9, -7)$.

- Find the slopes, side lengths, and diagonal lengths of the quadrilateral.
- Name the properties used to classify the quadrilateral.
- Classify the quadrilateral.

Question 6

Naomi is a landscape engineer. She is designing a garden on a coordinate grid using computer software. The vertices of the quadrilaterals in the garden are shown.

Quadrilateral <i>BRHW</i>	Quadrilateral <i>TFSN</i>
$B(3, 5)$, $R(5, 1)$, $H(0, -2)$, and $W(-2, 2)$	$T(9, 6)$, $F(13, 8)$, $S(16, 2)$, and $N(12, 0)$

- Find the slopes, side lengths, and diagonal lengths of each quadrilateral.
- Choose the quadrilateral that is a rectangle.
- Name the properties used.

Question 7

Quadrilateral $VWXY$ has vertices at $V(7, 9)$, $W(5, 5)$, $X(1, 3)$, and $Y(3, 7)$.

Provide the mathematics that will show that quadrilateral $VWXY$ is a rhombus, but not a square.

- Find the slopes, side lengths, and diagonal lengths of the quadrilateral.
- Name the properties that show why $VWXY$ is not a square.

Question 8

Quadrilateral $DSKM$ has vertices at $D(5, 8)$, $K(2, 5)$, and $M(2, 8)$.

Point S		
(4, 5)	(6, 3)	(7, 3)

- Determine the slope and length of $\overline{KD}$.
- Find the slope of $\overline{MS}$ using each point S .
- Where would vertex S need to be in order for quadrilateral $DSKM$ to be a kite?
- Verify that adjacent sides of $DSKM$ are congruent.